U23IT484 – CLOUD COMPUTING TECHNOLOGIES

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Roll No: 24AM113

ROUND 2 (i) – IMPLEMENT AWS STORAGE

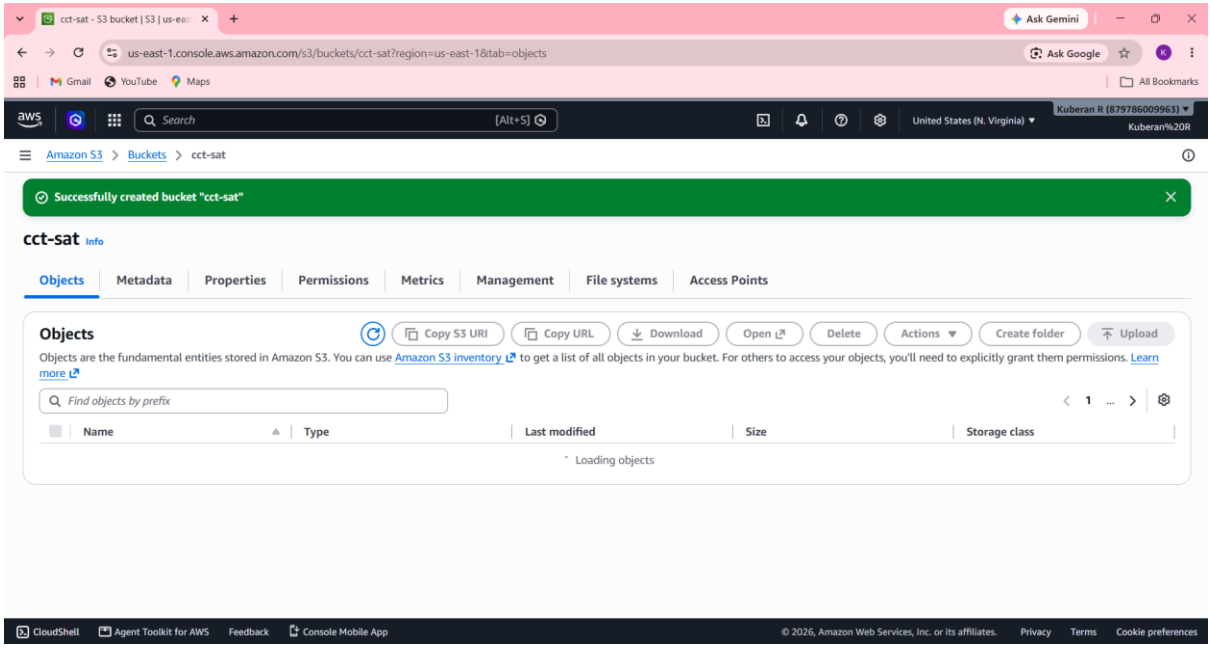
Objective

To implement secure cloud storage using **Amazon S3** with encryption, versioning, access control, and protection against accidental or unauthorized data deletion.

Procedure

Step 1 – Open Amazon S3

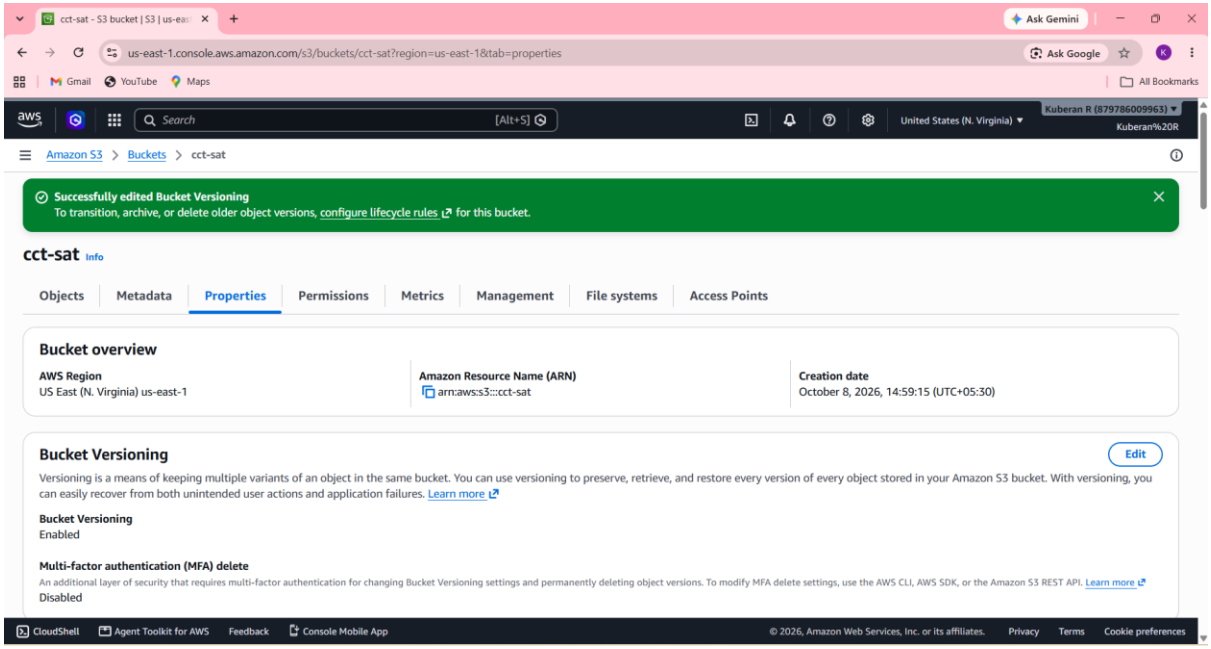
1. Sign in to the **AWS Management Console**.
2. Search for **S3** and open the Amazon S3 service.
3. Navigate to **Buckets** and select **Create bucket**.
4. Enter a globally unique bucket name.
5. Select the required AWS Region.
6. Keep **Block all public access** enabled.
7. Create the bucket.



Step 2 – Enable S3 Versioning

1. Open the newly created S3 bucket.
2. Go to the **Properties** tab.
3. Find **Bucket Versioning**.
4. Select **Enable**.
5. Save the configuration.

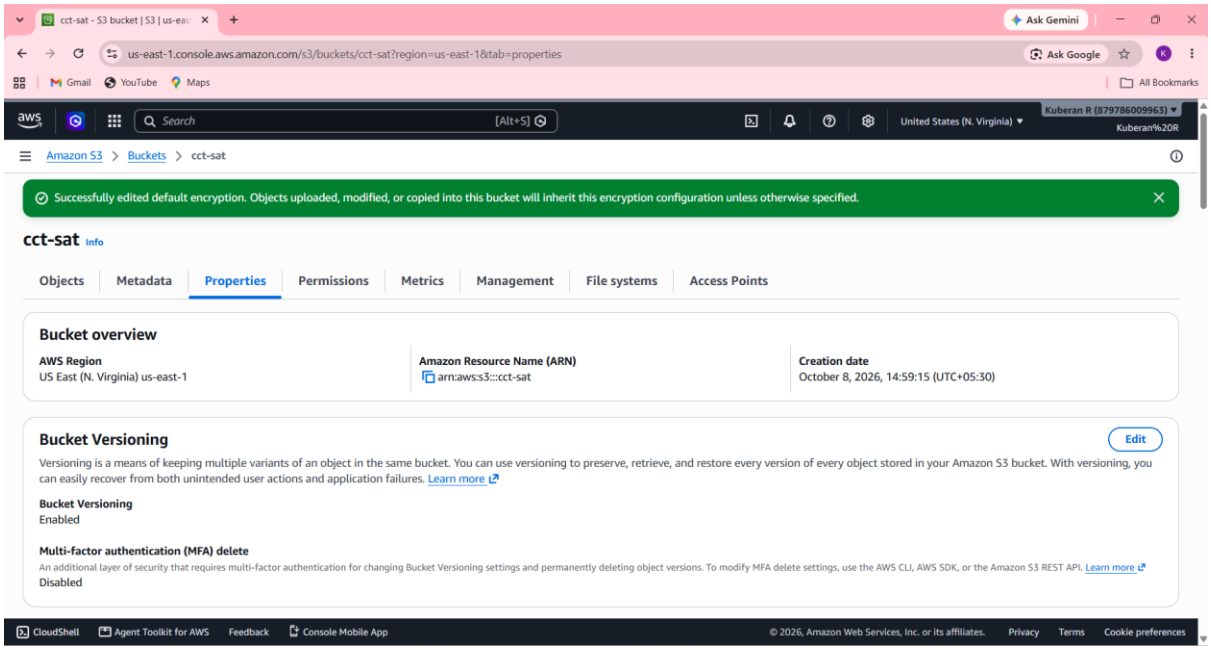
Versioning helps retain previous versions of objects and provides recovery from accidental overwrites or deletions.



Step 3 – Configure Server-Side Encryption

1. In the S3 bucket, open the **Properties** tab.
2. Locate **Default encryption**.
3. Enable server-side encryption.
4. Select **SSE-S3** for a simple student implementation.
5. Save the configuration.

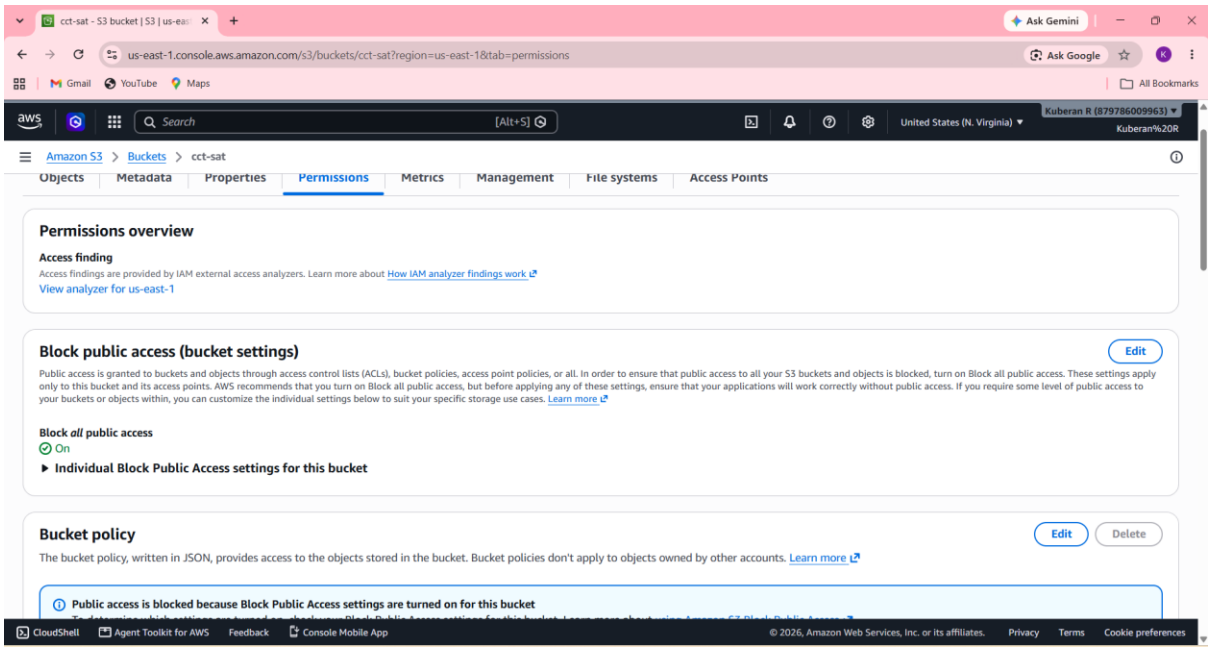
This ensures that objects stored in the bucket are encrypted at rest.



Step 4 – Configure Public Access Protection

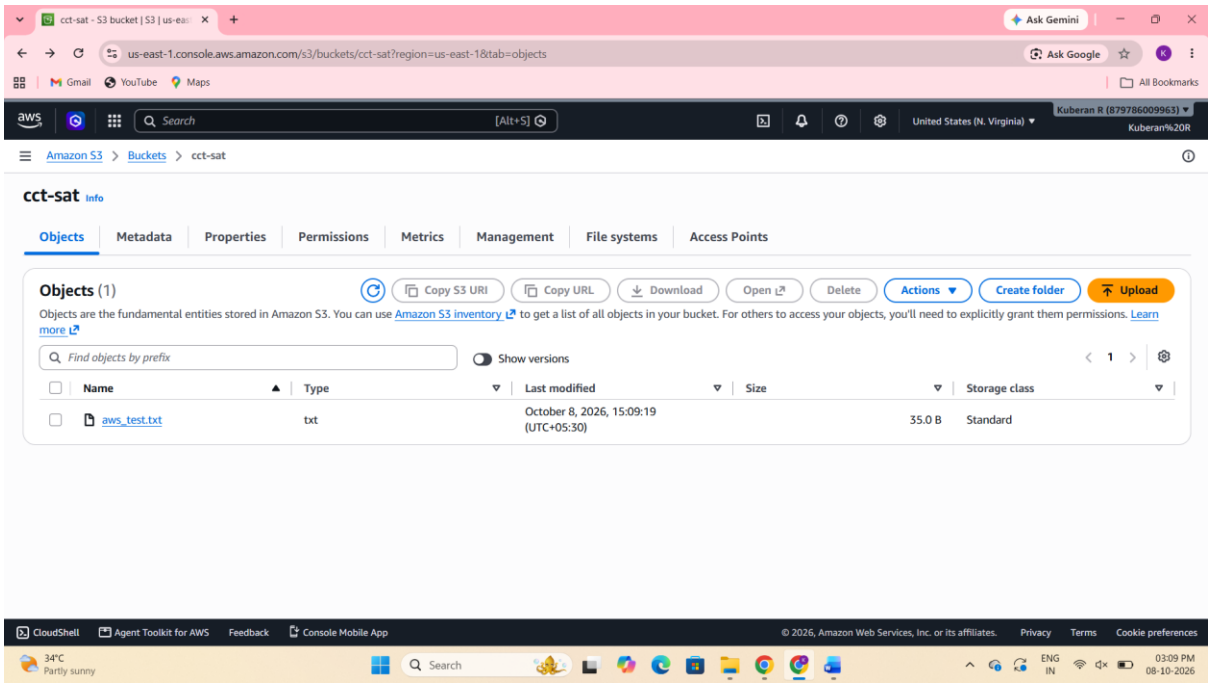
1. Open the bucket's **Permissions** tab.
2. Verify that **Block all public access** is enabled.
3. Confirm that no public bucket policy is configured.
4. Save the settings if any changes were made.

This prevents the storage bucket from accidentally becoming publicly accessible.



Step 5 – Upload a Test Object

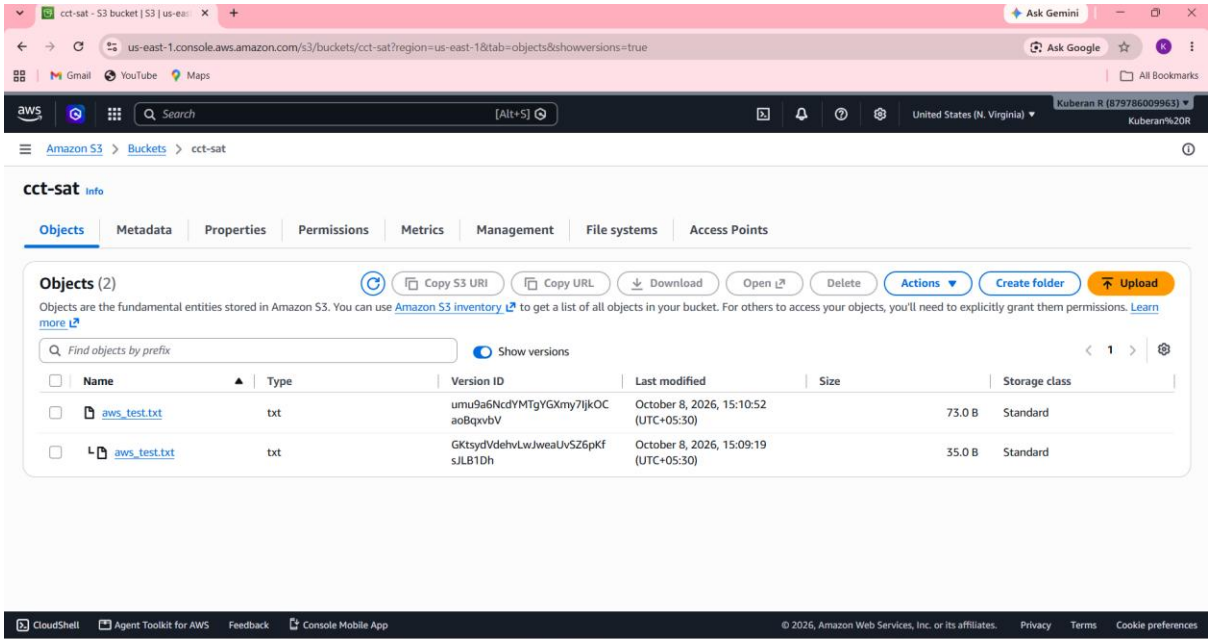
1. Open the S3 bucket.
2. Select **Objects → Upload**.
3. Upload a small test file.
4. Verify that the object appears in the bucket.
5. Open the object's Properties and verify its encryption/version information.



Step 7 – Test Versioning

1. Upload a file with the same name again after modifying its contents.
2. Open the bucket and enable **Show versions**.
3. Verify that multiple versions of the object are available.

This demonstrates how S3 versioning can help recover previous data.



ROUND 2 (ii) – BUILD END-TO-END AWS SOLUTION

Objective

To build an end-to-end AWS cloud solution by integrating **VPC, EC2, RDS, S3, IAM, and security controls** to provide a secure and functional application environment.

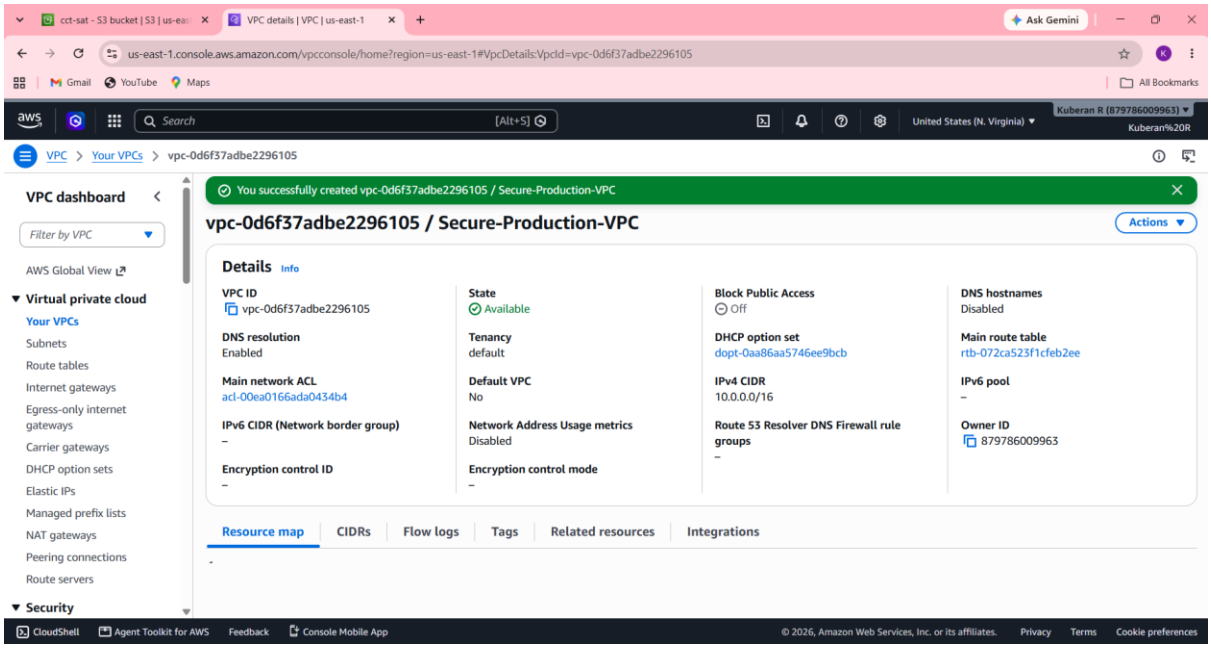
Procedure

Step 1 – Create the VPC

1. Open **AWS Console → VPC**.
2. Select **Create VPC**.
3. Choose **VPC only**.
4. Give the VPC a suitable name, such as Secure-Production-VPC.
5. Use a private CIDR range such as:

   10.0.0.0/16

6. Create the VPC.



Step 2 – Create the Subnets

Create separate subnets for the application and database components.

For example:

VPC: 10.0.0.0/16

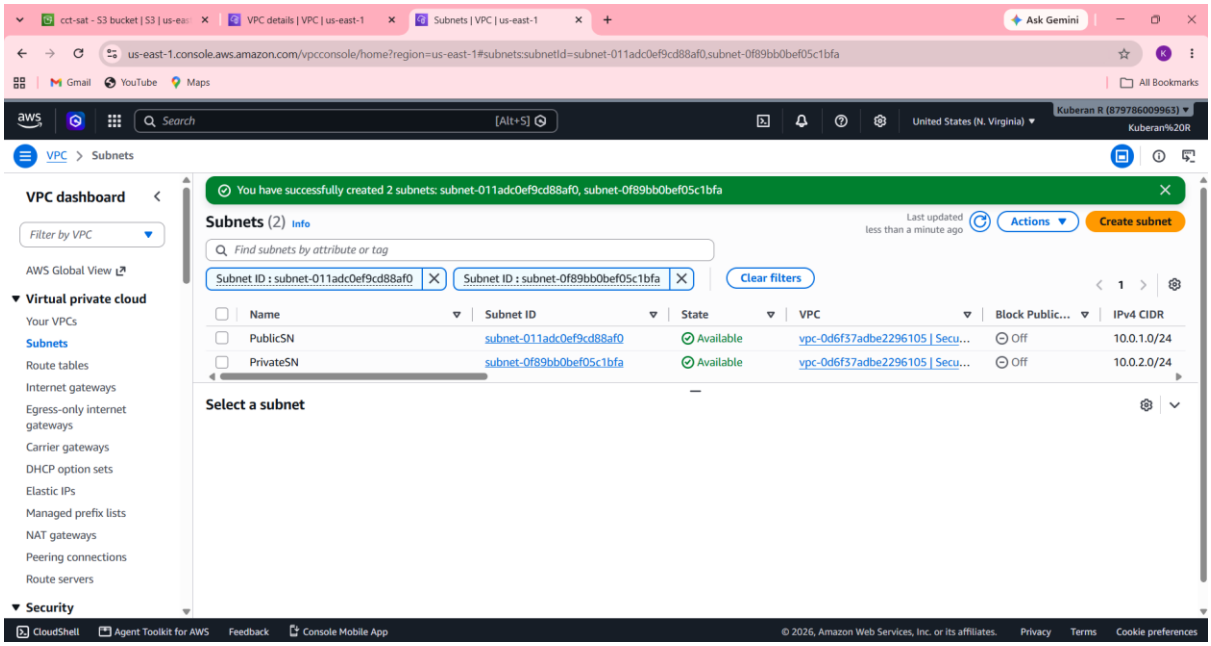
Public Subnet:

10.0.1.0/24

Private Subnet:

10.0.2.0/24

1. Open **VPC → Subnets**.
2. Create the public subnet.
3. Create the private subnet.
4. Associate both with the production VPC.



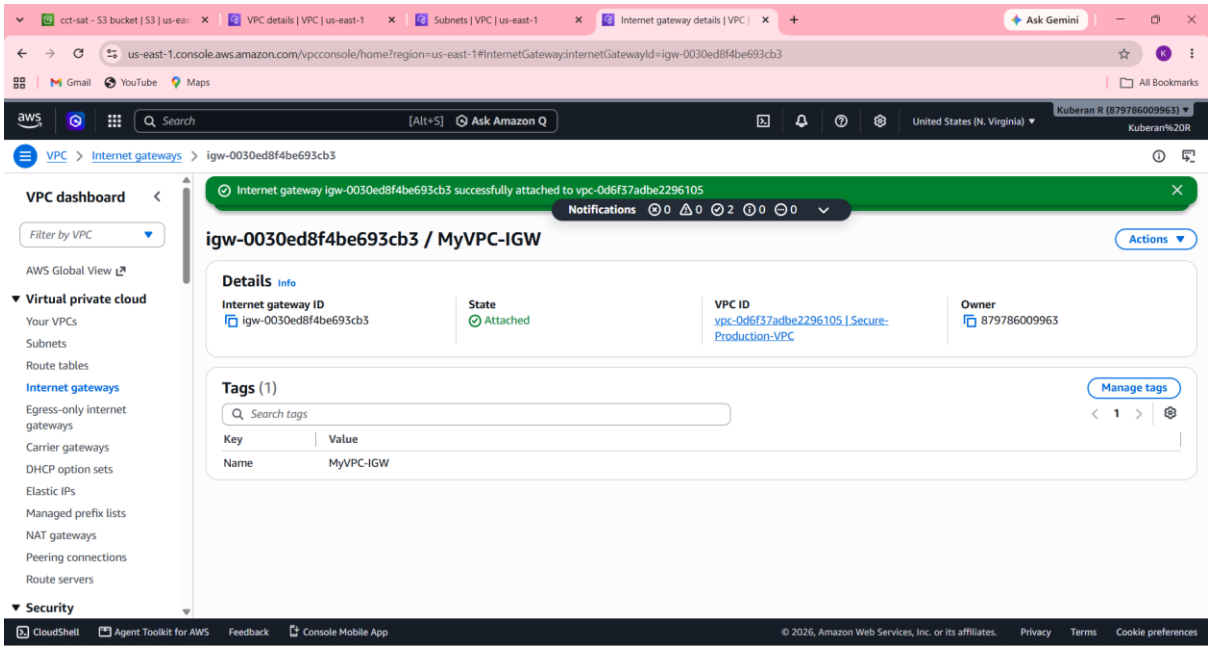
Step 3 – Configure Internet Connectivity

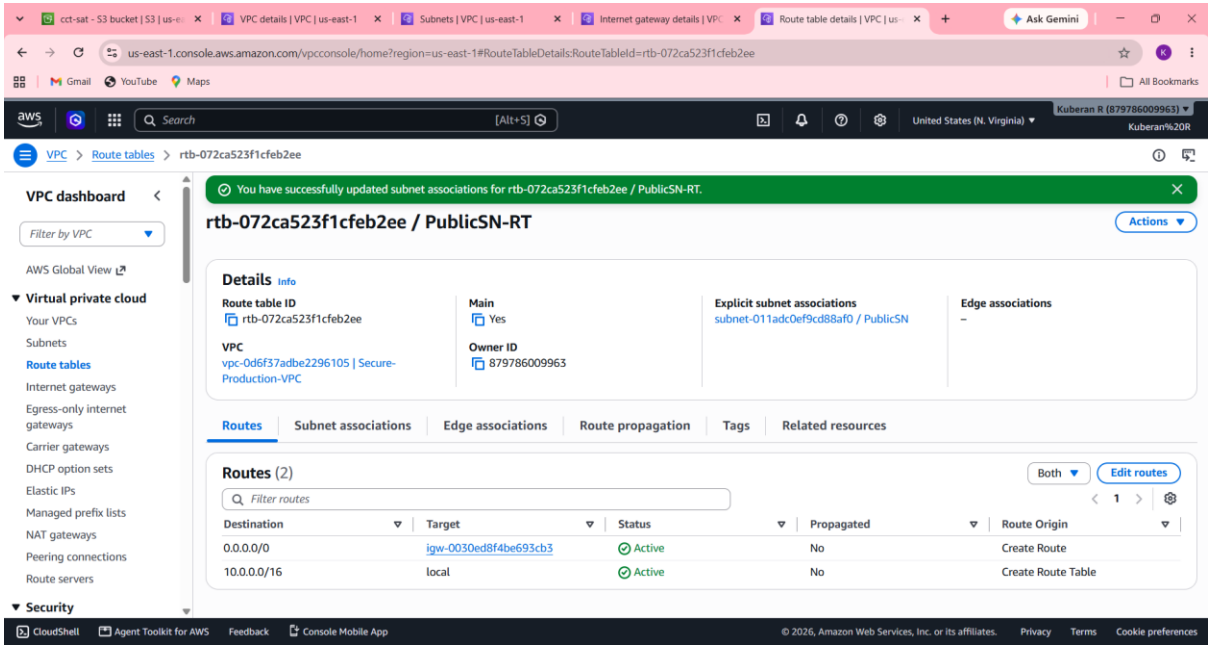
For the EC2 application server to receive required internet connectivity:

1. Create an **Internet Gateway**.
2. Attach it to the VPC.
3. Create/configure a route table for the public subnet.
4. Add the route:

0.0.0.0/0 → Internet Gateway

5. Associate the route table with the public subnet.





Step 4 – Create Security Groups

Create separate security groups for EC2 and RDS.

EC2 Security Group

Allow only the required application traffic.

For example:

SSH → Your IP only

HTTP → Required source

HTTPS → Required source

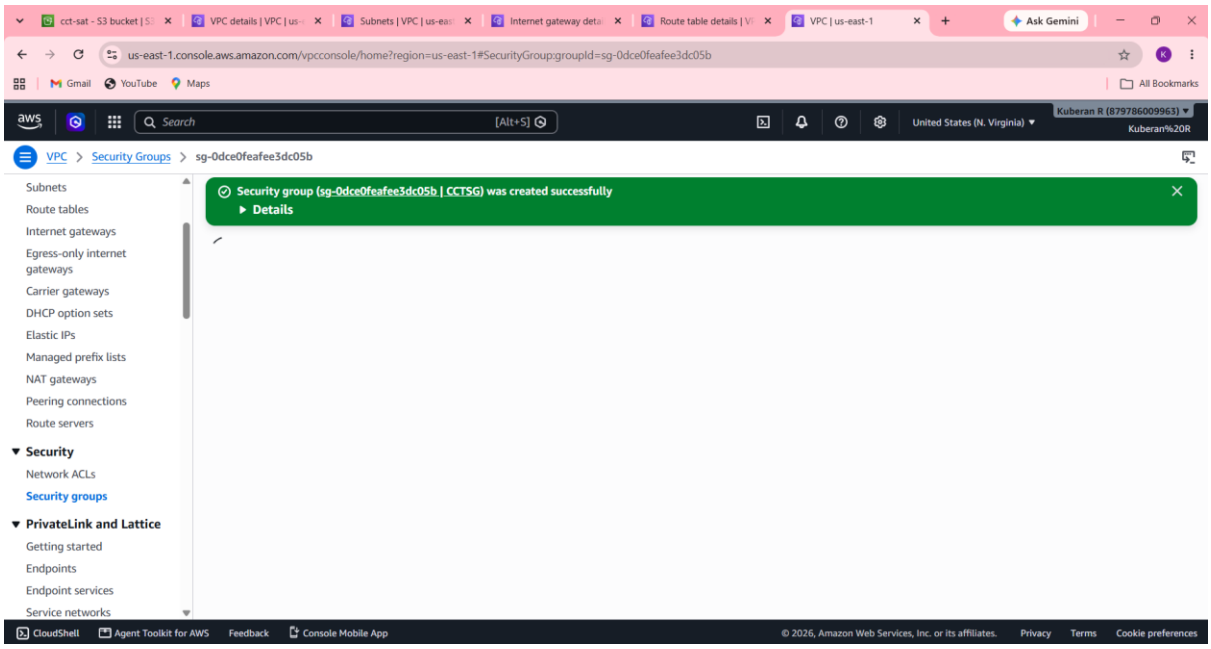
RDS Security Group

Do **not** allow database access from the entire internet.

Instead:

MySQL 3306 → EC2 Security Group

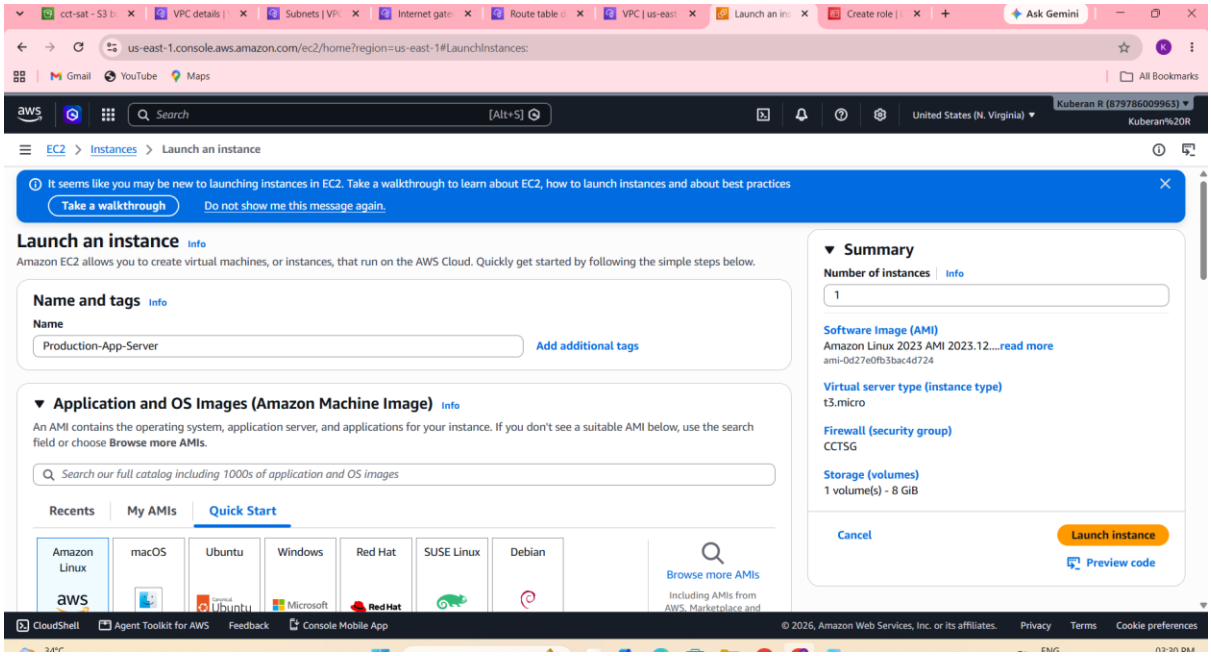
This means only the EC2 application server can communicate with the database.



Step 5 – Launch the EC2 Instance

1. Open **EC2 → Instances**.
2. Select **Launch Instance**.
3. Give it a name such as:

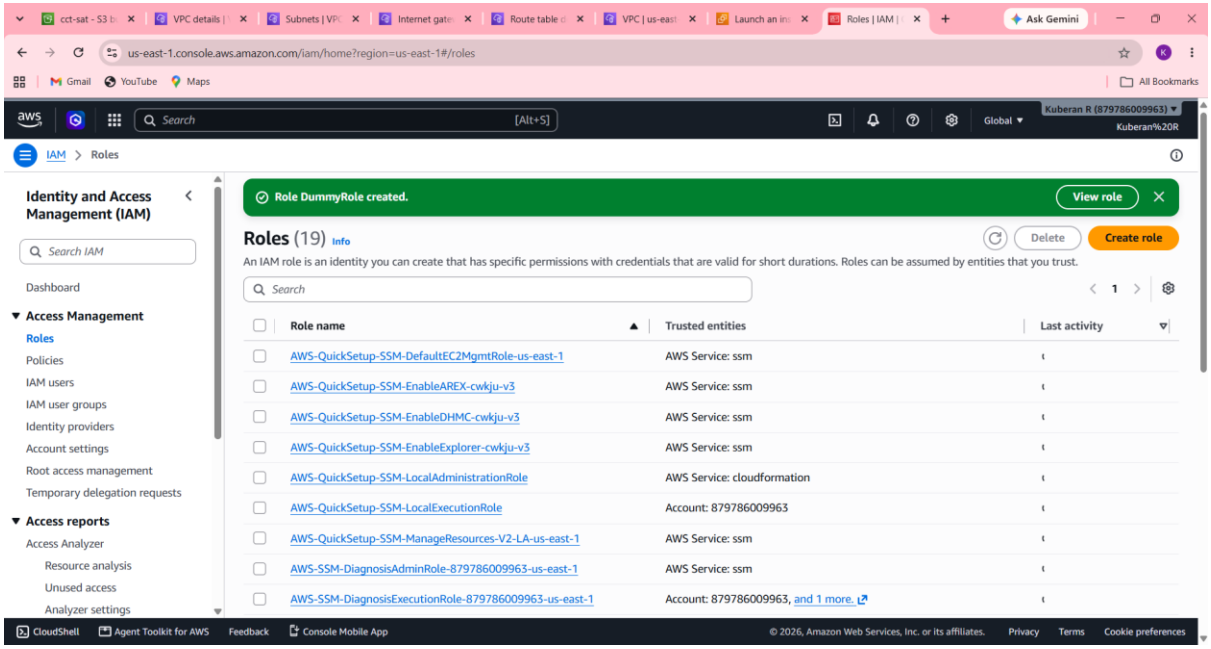
   Production-App-Server
4. Select an appropriate free-tier-eligible AMI/instance type if available to you.
5. Select the previously created VPC.
6. Select the **public subnet**.
7. Enable a public IP only if required for your implementation.
8. Select the EC2 security group created earlier.
9. Launch the instance.



Step 6 – Create an IAM Role for EC2

Instead of storing AWS access keys inside the EC2 machine:

1. Open **IAM → Roles**.
2. Create a role for **EC2**.
3. Attach a restricted S3 policy.
4. Allow only the required S3 actions.
5. Attach the role to the EC2 instance.



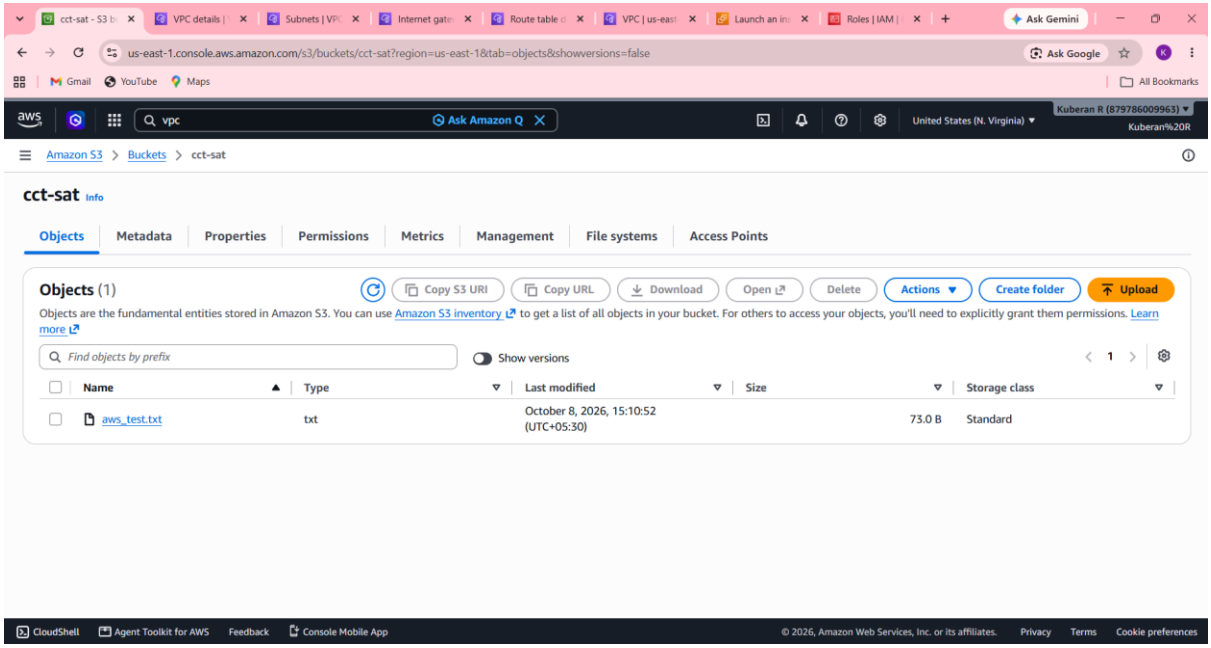
ROUND 3 – DATA INTEGRATION AND DEPLOYMENT

Objective

To integrate application data with AWS services and deploy the application using **EC2, RDS, and S3**, while validating secure communication, data access, monitoring, and deployment.

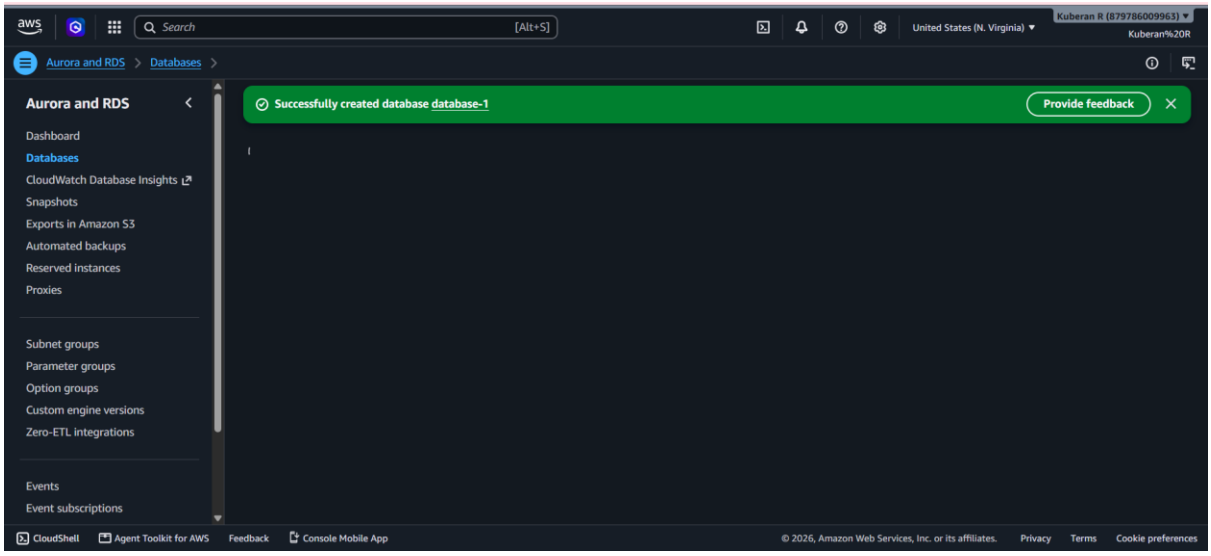
Step 1 – Upload Application Data to S3

1. Open the S3 bucket created in Round 2.
2. Create a suitable folder/prefix for application data if required.
3. Upload the required dataset/files.
4. Verify that the objects are encrypted and versioned.



Step 2 – Configure the RDS Database

1. Open the RDS database created in Round 2.
2. Connect to the database from the EC2 instance.
3. Create the required database/table structure.
4. Insert sample application data.
5. Verify that the data can be retrieved successfully.


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